AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI

ANNUAL EXAMINATIONS (THEORY) 2025

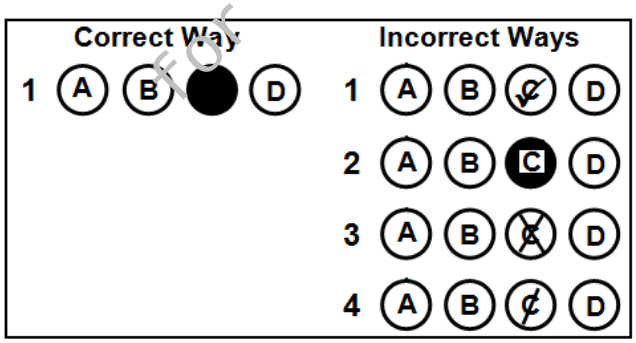
Mathematics Paper I

Time: 1 hour 30 minutes Marks: 50

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 50 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. A formulae list is provided on page 2 and 3. You may refer to it during the paper, if you wish.
8. You may use a scientific calculator if you wish.

List of Formulae

Note:

- The symbols used in the formulae have their usual meaning.

Complex Numbers

$|z| = \sqrt{a^2 + b^2}$

Matrices and Determinants

$A_{ij} = (-1)^{i+j} M_{ij}$ $\quad AdjA = (A_{ij})^t$ $\quad A^{-1} = \frac{1}{|A|} AdjA$

Sequence & Series and Miscellaneous Series

$a_n = a + (n-1)d$ $\quad A = \frac{a+b}{2}$ $\quad S_n = \frac{n}{2}\left(2a + (n-1)d\right)$

$a_n = ar^{n-1}$ $\quad G = \pm\sqrt{ab}$ $\quad H = \frac{2ab}{a+b}$

$S_n = \frac{a\left(1-r^n\right)}{1-r}$, if $|r| < 1$ $\quad S_n = \frac{a\left(r^n - 1\right)}{r-1}$, if $|r| > 1$ $\quad S_\infty = \frac{a}{1-r}$

$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$ $\quad \sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$ $\quad \sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2$

Permutations, Combinations and Probability

${}^nP_r = \frac{n!}{(n-r)!}$ $\quad {}^nC_r = \frac{n!}{(n-r)!r!}$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$ $\quad P(A/B) = \frac{P(A \cap B)}{P(B)}$ $\quad P(A \cap B) = P(A) \times P(B)$

Binomial Theorem and Mathematical Induction

$(a+x)^n = \binom{n}{0}a^n + \binom{n}{1}a^{n-1}x + \binom{n}{2}a^{n-2}x^2 + \binom{n}{3}a^{n-3}x^3 + \ldots + \binom{n}{n-1}a^1 x^{n-1} + x^n$

$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \ldots + \frac{n(n-1)(n-2)\ldots(n-r+1)}{r!}x^r + \ldots$

$T_{r+1} = \binom{n}{r}a^{n-r}x^r$

Quadratic Equation

$x^2 - Sx + P = 0$ $\quad x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ $\quad D = b^2 - 4ac$

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Introduction to Trigonometry and Trigonometric Identities

$l = r\theta$

$\sin^2\theta + \cos^2\theta = 1$

$1 + \tan^2\theta = \sec^2\theta$

$1 + \cot^2\theta = \operatorname{cosec}^2\theta$

$\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$

$\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$

$\tan(\alpha \pm \beta) = \dfrac{\tan\alpha \pm \tan\beta}{1 \mp \tan\alpha\tan\beta}$

$\cos\dfrac{\theta}{2} = \pm\sqrt{\dfrac{1+\cos\theta}{2}}$

$\sin\dfrac{\theta}{2} = \pm\sqrt{\dfrac{1-\cos\theta}{2}}$

$\tan\dfrac{\theta}{2} = \pm\sqrt{\dfrac{1-\cos\theta}{1+\cos\theta}}$

$\dfrac{a}{\sin\alpha} = \dfrac{b}{\sin\beta} = \dfrac{c}{\sin\gamma}$

$a^2 = b^2 + c^2 - 2bc\cos\alpha$

$\dfrac{a-b}{a+b} = \dfrac{\tan\frac{\alpha-\beta}{2}}{\tan\frac{\alpha+\beta}{2}}$

$\cos P - \cos Q = -2\sin\dfrac{P+Q}{2}\sin\dfrac{P-Q}{2}$

$\sin P - \sin Q = 2\cos\dfrac{P+Q}{2}\sin\dfrac{P-Q}{2}$

$\cos P + \cos Q = 2\cos\dfrac{P+Q}{2}\cos\dfrac{P-Q}{2}$

$\sin P + \sin Q = 2\sin\dfrac{P+Q}{2}\cos\dfrac{P-Q}{2}$

$\sin\dfrac{\alpha}{2} = \sqrt{\dfrac{(s-b)(s-c)}{bc}}$

$\tan\dfrac{\alpha}{2} = \sqrt{\dfrac{(s-b)(s-c)}{s(s-a)}}$

$\cos\dfrac{\alpha}{2} = \sqrt{\dfrac{s(s-a)}{bc}}$

Application of Trigonometry

$\Delta = \dfrac{1}{2}bc\sin\alpha = \dfrac{1}{2}ac\sin\beta = \dfrac{1}{2}ab\sin\gamma$

$\Delta = \sqrt{s(s-a)(s-b)(s-c)}$

$\Delta = \dfrac{c^2\sin\alpha\sin\beta}{2\sin\gamma} = \dfrac{b^2\sin\alpha\sin\gamma}{2\sin\beta} = \dfrac{a^2\sin\beta\sin\gamma}{2\sin\alpha}$

$R = \dfrac{a}{2\sin\alpha} = \dfrac{b}{2\sin\beta} = \dfrac{c}{2\sin\gamma}$

$r_1 = \dfrac{\Delta}{s-a}$, $r_2 = \dfrac{\Delta}{s-b}$ and $r_3 = \dfrac{\Delta}{s-c}$

$r = \dfrac{\Delta}{s}$

$R = \dfrac{abc}{4\Delta}$

Inverse Trigonometric Functions and Trigonometric Equations

$\sin^{-1}A \pm \sin^{-1}B = \sin^{-1}\left(A\sqrt{1-B^2} \pm B\sqrt{1-A^2}\right)$

$\cos^{-1}A \pm \cos^{-1}B = \cos^{-1}\left(AB \mp \sqrt{(1-A^2)(1-B^2)}\right)$

$\tan^{-1}A \pm \tan^{-1}B = \tan^{-1}\left(\dfrac{A \pm B}{1 \mp AB}\right)$

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1. The simplified form of $2i \div (1+i)$ is equal to

   A. $1+i$.
   B. $1-i$.
   C. $2+2i$.
   D. $2-2i$.

2. If $Z = a - ib$ and $|Z| = 2$, then the relationship between a and b is

   A. $a^2 - b^2 = 4$
   B. $a^2 + b^2 = 4$
   C. $a + b = 2$
   D. $a - b = 2$

3. If $z = a - ib$, then which one of the following options is false?

   A. $|z| = |-z|$
   B. $|z| = |\bar{z}|$
   C. $|z| = -|\bar{z}|$
   D. $|z| = |-\bar{z}|$

4. The expression $x^4 + y^4$ can be factorised as

   A. $(x^2 + iy^2)(x^2 + iy^2)$
   B. $(x^2 - iy^2)^2 + 2ix^2y^2$
   C. $(x^2 - iy^2)(x^2 + iy^2)$
   D. $(x^2 + iy^2)^2 - 2ix^2y^2$

5. If A = $\begin{bmatrix} 1 & 5 \\ 5 & 6 \end{bmatrix}$ is a skew-symmetric matrix, then A^t is equal to

   A. $\begin{bmatrix} 1 & 5 \\ 5 & 6 \end{bmatrix}$.
   B. $\begin{bmatrix} -1 & 5 \\ 5 & -6 \end{bmatrix}$.
   C. $\begin{bmatrix} 6 & -5 \\ -5 & 1 \end{bmatrix}$.
   D. $\begin{bmatrix} -1 & -5 \\ -5 & -6 \end{bmatrix}$.

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6. The product of matrices $\begin{bmatrix} 3 & 2 \\ 4 & 5 \end{bmatrix} \times \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix}$ will be equal to

A. $k^2 \begin{bmatrix} 3 & 2 \\ 4 & 5 \end{bmatrix}$.

B. $\begin{bmatrix} 3k & 0 \\ 0 & 5k \end{bmatrix}$.

C. $k \begin{bmatrix} 3 & 2 \\ 4 & 5 \end{bmatrix}$.

D. $\begin{bmatrix} 0 & 2k \\ 4k & 0 \end{bmatrix}$.

7. For the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & x \\ 2 & 3 & 1 \end{bmatrix}$, the minor element at position of x

A. is -1
B. is 1
C. is 7
D. cannot be found

8. The determinant of the matrix $A = \begin{bmatrix} 1 & -1 & 0 \\ 2 & -1 & 1 \\ 3 & -2 & -2 \end{bmatrix}$ is

A. -7
B. -3
C. 7
D. 11

9. If $A = \begin{bmatrix} p & 1 & -p \\ 2 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ is a singular matrix, then the value of p will be

A. -1
B. 0
C. 1
D. 2

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10. If $\begin{vmatrix} a & b & c \\ b & a & b \\ c & c & a \end{vmatrix} = x$, then $\begin{vmatrix} a & -3b & c \\ b & -3a & b \\ c & -3c & a \end{vmatrix}$ will be equal to

A. 0
B. 1
C. $3x$
D. $-3x$

11. The determinant of $\begin{vmatrix} 3 & 6 & 3m \\ 1 & 2 & m \\ ? & ? & ? \end{vmatrix}$

A. is 3
B. is 1
C. is 0
D. cannot be determined

12. It is given that $\begin{vmatrix} 5 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{vmatrix} = 5$, $\begin{vmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ -1 & 0 & 1 \end{vmatrix} = 2$, $\begin{vmatrix} 5 & 1 & 1 \\ 0 & 0 & 2 \\ 0 & -1 & 1 \end{vmatrix} = 10$ and $\begin{vmatrix} 5 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{vmatrix} = -5$.

On solving the equation $\begin{bmatrix} 5 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$, the value of y will be

A. -1
B. $\frac{2}{5}$
C. $\frac{1}{2}$
D. 2

13. The general term of the sequence $a^2, -a^3, a^4, -a^5, \ldots$ is

A. $-\frac{a^n}{a}$
B. $\frac{(-a)^n}{a}$
C. $-a^{n+1}$
D. $(-a)^{n+1}$

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14. If the first term of an arithmetic sequence is 3 and the common difference is 5, then the fourth term is

    A. 12
    B. 15
    C. 18
    D. 23

15. The sum of the first three terms of an arithmetic series is 18. If the common difference of the series is 3, then the first term of the series will be

    A. 3
    B. 5
    C. 9
    D. 15

16. At the end of the first week, the plant's height is 1.67 cm, and its growth rate for each subsequent week is 4% higher than the previous week.

    The expression that calculates the height of the plant at the end of 9 weeks can be written as

    A. $1.67 \times \left(\frac{4}{100}\right)^8$
    B. $1.67 \times \left(\frac{104}{100}\right)^8$
    C. $1.67 \times \left(\frac{4}{100}\right)^9$
    D. $1.67 \times \left(\frac{104}{100}\right)^9$

17. Consider the sequence $1, 1\times 3, 1\times 3^2, 1\times 3^3, \ldots$.

    The sum of the first 10 terms of the given sequence is

    A. 243
    B. 5904
    C. 9841
    D. 29524

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18. The harmonic mean between $\frac{1}{2}$ and $\frac{1}{3}$ is

A. $\frac{2}{5}$

B. $\frac{5}{12}$

C. $\frac{5}{2}$

D. $\frac{12}{5}$

19. If $\sum_{k=1}^{2} k = 3$, then the value of $\sum_{k=3}^{n} k$ is equal to

A. $\frac{n(n+1)}{2} - 3$

B. $n\left(n+\frac{1}{2}\right)-3$

C. $3 - \frac{n(n+1)}{2}$

D. $3 - n\left(n+\frac{1}{2}\right)$

20. Which of the following options is equivalent to $\sum_{1}^{n}\left(2n^2 - n + 2\right)$?

A. $2\sum_{1}^{n} n^2 - \sum_{1}^{n} n + 2$

B. $2\sum_{1}^{n} n^3 - \sum_{1}^{n} n^2 + 2$

C. $2\sum_{1}^{n} n^2 - \sum_{1}^{n} n + 2n$

D. $2\sum_{1}^{n} n^3 - \sum_{1}^{n} n^2 + 2n$

21. The value of $1! - \left[0! \times \frac{n!}{(n-1)!}\right]$ is equal to

A. 0
B. 1
C. $1-n$
D. $(1-n)!$

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22. In a garden, five chairs are placed in a row.

The expression to find the number of ways these chairs can be filled by five persons, when two persons always sit next to each other, is

A. $5!$
B. $4!$
C. $2! \times 4!$
D. $2! \times 5!$

23. On evaluating, the combination $\binom{ab}{ab}$ reduces to

A. 1
B. ab
C. $(ab)!$
D. $a!b!$

24. If A and B are two events such that $P(A) = 0.25$ and $P(B \mid A) = 0.75$, then $P(A \cap B)$ is equal to

A. 0.1875
B. 0.3333
C. 0.5
D. 1

25. In a class of 40 students, 15 are to be selected for the art society. There are 3 particular students who are good in arts and the selectors desire to include them in the society.

In how many ways the society members can be selected?

A. $\binom{40}{15}$
B. $\binom{37}{15}$
C. $\binom{37}{12}$
D. $\binom{40}{12}$

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26. If $P(M \cap N) = \frac{m}{c}$, then the value of $P(M^c \cup N^c)$ is equal to

(**Note**: $m < c$)

A. 0
B. 1
C. $\frac{c}{m}$
D. $1 - \frac{m}{c}$

27. $1^3 + 2^3 + 3^3 + ... + K^3 = \frac{K^2(K+1)^2}{4}$ is true for $(K+1)^{th}$ term, the term that should be added to both sides is

A. $K^3 + 1$
B. $(K+1)^3$
C. $\frac{(K+1)^2(K+2)^2}{4}$
D. $\frac{(K+1)^3(K+2)^3}{4}$

28. The value of p in the binomial expansion $(1+px)^3 = 1 + 12x + 48x^2 + 64x^3$ is

A. 3
B. 4
C. 8
D. 12

29. In the binomial expansion of $\left(x - \frac{y}{4}\right)^5$, the sum of the coefficient of first and second term is

A. $1 + {}^5C_1\left(\frac{1}{4}\right)$.
B. $1 - {}^5C_1\left(\frac{1}{4}\right)$.
C. ${}^5C_1 + {}^5C_2\left(\frac{1}{4}\right)$.
D. ${}^5C_1 - {}^5C_2\left(\frac{1}{4}\right)$.

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30. The value of *b* in the expansion of $(1-x)^{-5} = 1+5x+bx^2+\ldots$, will be

A. -15
B. -10
C. 10
D. 15

31. What should be added to make $x^2 - x + \frac{3}{4}$ a perfect square?

A. $-\frac{1}{4}$
B. $-\frac{1}{2}$
C. $\frac{1}{4}$
D. $\frac{1}{2}$

32. To reduce the exponential equation $3x^4 + 2x^2 + 5 = 0$ to a quadratic form in terms of *y*, we substitute

A. $x = y$
B. $x = y^2$
C. $x^2 = y$
D. $x^2 = y^2$

33. The discriminant of $2x^2 - 6x - 3 = 0$ is

A. -60
B. -12
C. 12
D. 60

34. The value of $\sqrt{(\omega)^3 - (-1-\omega-\omega^2)}$ is equal to

A. 0
B. 1
C. $\sqrt{2}$
D. 4

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35. If sum of the roots of $ax^2 + bx + c = 0$ is square of the product of its roots, then the CORRECT condition will be

A. $c^2 = ab$
B. $b^2 = ac$
C. $b^2 = -ac$
D. $c^2 = -ab$

36. If $\frac{1}{\alpha}$ and β are the roots of a quadratic equation, then the equation will be

(**Note**: $\alpha \neq 0$)

A. $x^2 + (1 + \alpha\beta)x + \frac{\beta}{\alpha} = 0$
B. $x^2 - (1 + \alpha\beta)x + \frac{\beta}{\alpha} = 0$
C. $x^2 - \left(\frac{1}{\alpha} + \beta\right)x + \frac{\beta}{\alpha} = 0$
D. $x^2 + \left(\frac{1}{\alpha} + \beta\right)x + \frac{\beta}{\alpha} = 0$

37. If $\tan x = \frac{3}{4}$, then $\cos x$ will be equal to

A. $\pm\frac{\sqrt{7}}{4}$
B. $\pm\frac{4}{\sqrt{7}}$
C. $\pm\frac{4}{5}$
D. $\pm\frac{5}{4}$

38. If $\sin\theta = 2\cos\theta$, then $1 + \cot\theta$ is equal to

A. $\frac{3}{2}$
B. $\frac{2}{3}$
C. 2
D. 3

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39. On simplification of $\sec^2\theta - \tan^2\theta - 1$, we get

A. 0
B. 1
C. $2\tan^2\theta$
D. $2\sec^2\theta$

40. The distance between the points (0, 0) and $(\sin\alpha, -\cos\beta)$ is

A. 1
B. $\sin\alpha - \cos\beta$
C. $\sqrt{\sin^2\alpha - \cos^2\beta}$
D. $\sqrt{\sin^2\alpha + \cos^2\beta}$

41. Which of the following formulae is/ are equal to $\cos\theta$?

I. $\sin\left(\frac{\pi}{2} - \theta\right)$

II. $\sin\left(\frac{\pi}{2} + \theta\right)$

III. $\sin\left(\frac{3\pi}{2} - \theta\right)$

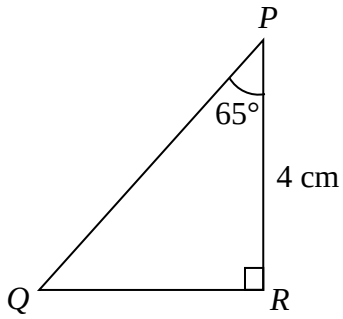
A. I only
B. III only
C. I and II
D. I and III

42. If $\sin\alpha = a$ and $\cos\alpha = b$, then the value of $\sin 2\alpha$ is equal to

A. ab
B. $\frac{a}{b}$
C. $\frac{2a}{b}$
D. $2ab$

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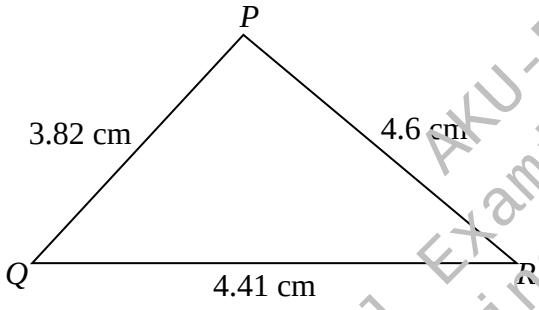
43. In the triangle *PQR*, the length of *PQ* is



NOT TO SCALE

A. 1.69 cm.
B. 3.62 cm.
C. 4.41 cm.
D. 9.46 cm.

44. In the triangle *PQR*, the measure angle of *P* is

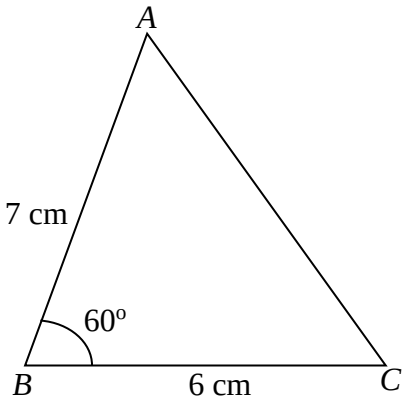


NOT TO SCALE

A. 50.12°
B. 62.36°
C. 67.52°
D. 68.49°

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45. In the given diagram, the area of a triangle *ABC* is



NOT TO SCALE

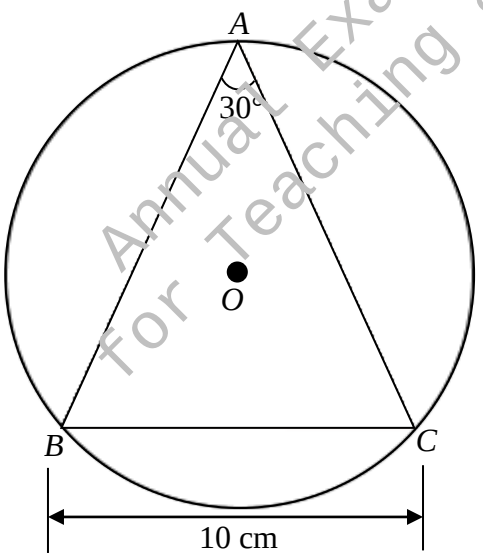
A. $21\ \text{cm}^2$

B. $\frac{21}{2}\ \text{cm}^2$

C. $21\sqrt{3}\ \text{cm}^2$

D. $\frac{21\sqrt{3}}{2}\ \text{cm}^2$

46. A circum-circle is drawn with a centre O as shown in the diagram. The triangle *ABC* is an isosceles triangle with $BC = 10$ cm and $\angle A = 30^\circ$.



NOT TO SCALE

The value of twice of circum-radius, i.e., 2*R* will be equal to

A. 5 cm.

B. 20 cm.

C. 25 cm.

D. 40 cm.

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47. The period of $\sec 8x$ is

A. $\frac{\pi}{4}$

B. $\frac{\pi}{2}$

C. 2π

D. 16π

48. The equation $a\tan^{-1}(b) = \sin^{-1}(c)$ will become $\tan\left(\frac{x}{a}\right) = b$, when $\sin^{-1}(c)$ is substituted by

A. x

B. ax

C. $\frac{a}{x}$

D. $\frac{x}{a}$

49. The solution set of $\sin\theta\cos\theta = 0$ is

A. $\left\{\frac{\pi}{2}, \pi\right\}$.

B. $\{\frac{n\pi}{2} + n\pi, n \in Z\}$.

C. $\{\frac{\pi}{2} + 2n\pi, n \in Z\} \cup \{2n\pi, n \in Z\}$.

D. $\{\frac{n\pi}{2} + 2n\pi, n \in Z\} \cup \{n\pi, n \in Z\}$.

50. The general solution of $\sin x - 1 = 0$ is

A. $\frac{\pi}{2} + 2n\pi$.

B. $\frac{\pi}{2} + n\pi$.

C. $\pi + 2n\pi$.

D. $\pi + n\pi$.

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